Akoizo Original Practice - Form 2 - Answer key

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which part of the vehicle is used to measure Vehicle Distance?

Unit: Planning the path: distances, geometry, and the reverse.
Worked-case data: Target Distance 7.00 m, Bottle Line Distance 2.00 m, vehicle length 50 cm, bottle 6.5 cm, desired margin 3 cm. How far does the MP travel forward and backward?
Separate investigation observations:
Setup: Target 8.00 m, Bottle Line 1.00 m, vehicle length 45 cm.
Run: MP forward travel programmed as 8.60 m.
Result: Bottle ended 1 cm short of the line.

A. The front axle

B. The center of mass

C. The Measurement Point at the rear

D. The bottle pusher

Correct: C

2. (2 points) How must the vehicle be started?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. With an unsharpened #2 pencil using a vertical motion

B. By hand push

C. With a phone app

D. By releasing a held wheel

Correct: A

3. (2 points) What happens if the vehicle tips the bottle over while pushing?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. The Bottle Distance is set to 400.0 cm

B. A -20 bonus is still earned

C. The run is restarted

D. Nothing, if it crosses the line

Correct: A

4. (2 points) Mounting the encoder on the motor shaft instead of the wheel axle with a 5:1 reduction:

A. Gives five times as many counts per wheel turn

B. Gives one-fifth as many counts

C. Has no effect

D. Prevents reversing

Correct: A

5. (4 points) Why is a centimeter of vehicle error more costly than a centimeter of bottle error?

answer: Vehicle Distance is scored at 2.0 points per cm while Bottle Distance is 1.0 point per cm, so the same error costs twice as much for the vehicle. (4 points).

6. (4 points) Why are large capacitors banned even though they store electrical energy?

answer: They could release energy faster than the batteries can supply instantaneously, giving power beyond the AA limit. (4 points).

7. (4 points) Explain how overshoot can cost points.

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

answer: Passing the target and correcting back adds time, may leave the MP off target, and any oscillation is counted in Run Time. (4 points).

8. (4 points) Explain the tradeoff in choosing how far past the line to push the bottle.

answer: More margin protects the -20 bonus but adds 1 point per cm of Bottle Distance; too little risks missing the line and losing the bonus. (4 points).

9. (4 points) Name two ways a run becomes a Failed Run.

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

answer: The vehicle travels the wrong direction, or starts before the supervisor is ready; also if it cannot be measured or is pushed. (4 points).

10. (4 points) Why should teams test with dented or off-weight bottles?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

answer: Real bottles vary from 80 to 100 g and may not be perfectly round, and the pusher must handle them. (4 points).

11. (6 points) Solve this case and explain your reasoning: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?

reasoning: Counts per wheel revolution = 100 x 3 = 300.
Distance per count = π x 6.5 ÷ 300 about 0.068 cm.
Counts for 860 cm = 860 ÷ 0.068 about 12,630. (3 points).

conclusion: Command about 12,630 counts, then correct the factor with a measured calibration run. (3 points).

12. (2 points) Which run scored lower (better)?

Supplied synthetic observations:
Pusher: Widest opening 20.0 cm.
Run 1: MP 6.0 cm off, bottle 5.0 cm past the line, time error 3.0 s.
Run 2: MP 3.0 cm off, bottle 8.0 cm short of the line, time error 0.5 s.

A. Run 1

B. Run 2

C. They tie

Correct: A

13. (2 points) What is the key test for whether a component breaks the energy-storage rule?

Supplied synthetic observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

A. Whether it is electronic

B. Whether it changes voltage

C. Whether it lets the vehicle exceed the instantaneous energy available from the batteries

Correct: C

14. (2 points) Define “Open-loop control” in the context of Electric Vehicle.

answer: Commands without measuring the result. (2 points).

15. (2 points) Define “Bottle Line Distance” in the context of Electric Vehicle.

answer: Target Point to Bottle Line, 1.00-3.00 m. (2 points).

16. (2 points) Define “Event time” in the context of Electric Vehicle.

answer: Eight minutes to adjust and make two runs. (2 points).

17. (2 points) Define “Funnel (V) entry” in the context of Electric Vehicle.

answer: Angled guides that center an offset bottle. (2 points).

18. (2 points) Define “Gear ratio” in the context of Electric Vehicle.

answer: Motor turns per wheel turn. (2 points).